DATA ANALYST

Muhammad Fauzan Rizkiatama Syam

Github Linkedin

DATA ANALYST



Muhammad Fauzan Rizkiatama Syam

PROFILE

- A Planology graduate transitioning into Data Analytics.
- Interested in data-driven decision-making and business insights.
- Aspiring to become a Data Analyst, with a long-term interest in renewable energy industry.

Education

- Bachelor's Degree in Urban and Regional Planning
- Universitas Islam Bandung (UNISBA)
- Graduated in 2023
- Academic focus: Spatial analysis and regional planning.

Professional Development & Projects

- Currently developing Data Analytics skills through the Dibimbing Data Analyst Bootcamp.
- Working with Python, PostgreSQL, Power BI, and Tableau.
- Completed hands-on projects involving customer segmentation, exploratory data analysis (EDA), and interactive dashboards.
- Focused on turning data into actionable business insights.

Github

Linkedin

Portofolio

Target Role

Data Analyst

Goals

Showcase project end-to-end data analysis.

Customer Segmentation & Behavioral Analysis: Driving Data-Driven Business Insights

Link Drive Project

Customer Churn Risk & Retention Analysis

Link Project

Main Project

Customer Churn Risk & Retention Analysis

Identifying Churn Risk, Revenue Exposure, and Customer Retention Strategies

Link Project

PROBLEM STATEMENT & BUSINESS QUESTIONS

Business Understanding

Problem Statement

- The company faces the risk of losing customers and revenue due to churn.
- Not all customers have the same business value or level of churn risk.
- The company needs to identify priority customers and develop more targeted retention strategies.

Business Questions

- Q1: Which high-value customers have a high risk of churning?
- Q2: What customer value and monthly revenue are at risk?
- Q3: Which customer characteristics are associated with churn?
- Q4: What retention strategies should be used, and how should the budget limit be determined?

BUSINESS UNDERSTANDING

Customer Identification & Analysis Goals

Customers are divided into four segments based on CLTV and churn score.

Segment	Characteristics
Priority 1	**High Value – High Risk**
Priority 2	High Value – Low Risk
Priority 3	Low Value – High Risk
Priority 4	Low Value – Low Risk

Analysis Goals

- Identify customers who should receive retention support first.
- Calculate the customer value and revenue exposed to churn risk.
- Analyze churn patterns by contract, tenure and customer satisfaction.
- Develop retention strategies and simulate a possible retention budget.

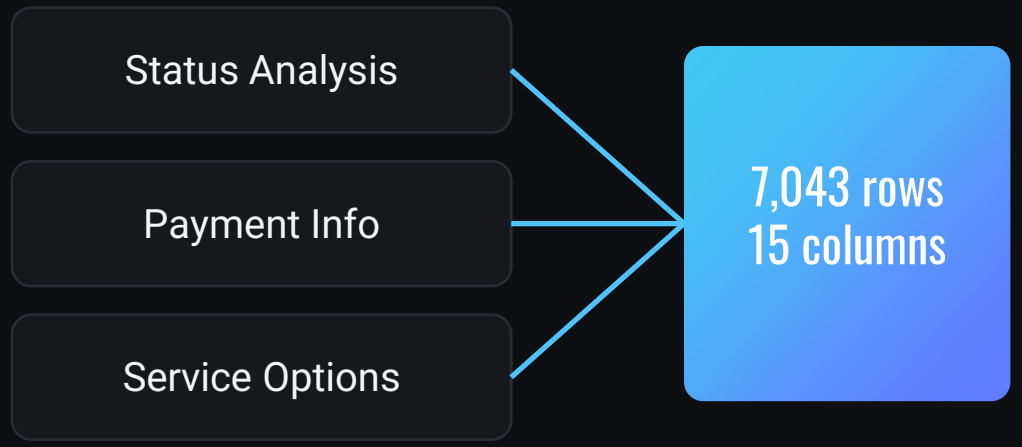
DATA PREPROCESSING

Variables & Data Integration

Required Variables

- Identity and status: customer_id, customer_status, churn_label, churn_value, churn_reason
- Value and risk: cltv, churn_score
- Contract and payment: contract, monthly_charges, payment_method
- Services and experience: tenure, internet_service, satisfaction_score, offer

Three tables combined into one dataset



Integration check

No missing customer_id values were found during either additional join.

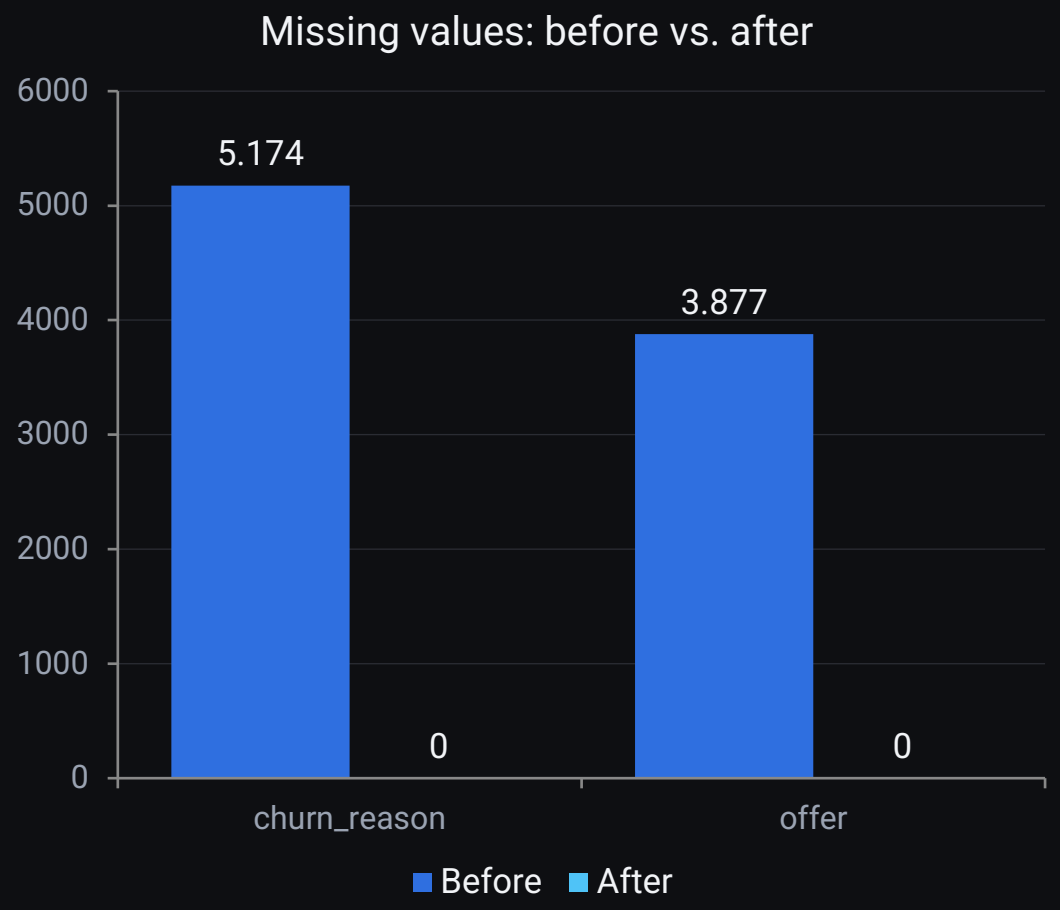
DATA PREPROCESSING

Missing Values

Column	Missing	Percentage
churn_reason	5,174	73.46%
offer	3,877	55.05%

Handling Missing Values

- churn_reason: replaced with Not Applicable.
- offer: replaced with None.
- CLTV, churn score, customer status, contract and monthly charges had no missing values after preprocessing.



DATA PREPROCESSING

Data Validation & Cleaning

Duplicate Check

- No duplicate rows were found in the dataset.

Standardization

- Cleaned column names and removed extra spaces.
- Checked consistency of churn_label and churn_value.
- Ensured customer status categories were consistent.

Data Types

- Categorical columns converted to the category type.
- Numerical columns kept in numerical formats for calculations.

Outlier Check (IQR)

- Checked CLTV, churn score, monthly charges and tenure.
- Zero outliers detected across all four variables.
- No records were removed.

DATA PREPARATION

Feature Engineering

Customer Tier

- Priority 1: High Value – High Risk
- Priority 2: High Value – Low Risk
- Priority 3: Low Value – High Risk
- Priority 4: Low Value – Low Risk

Tenure Group

- ≤ 1 year: New
- 1–2 years: Mid-term
- 2–4 years: Established
- > 4 years: Loyal

Thresholds

- Median CLTV: $4,527
- Median churn score: 61

CLTV

Estimates customer value over the subscription relationship.

Churn Score

Risk of leaving; a higher score means higher churn risk.

Projected Annual Charges

Estimates 12 months of charges at the current monthly rate.

Retention Budget Cap

Simulates a budget limit of 10% of CLTV.

Note: CLTV and churn score are already in the dataset. Median values are used only as segmentation thresholds.

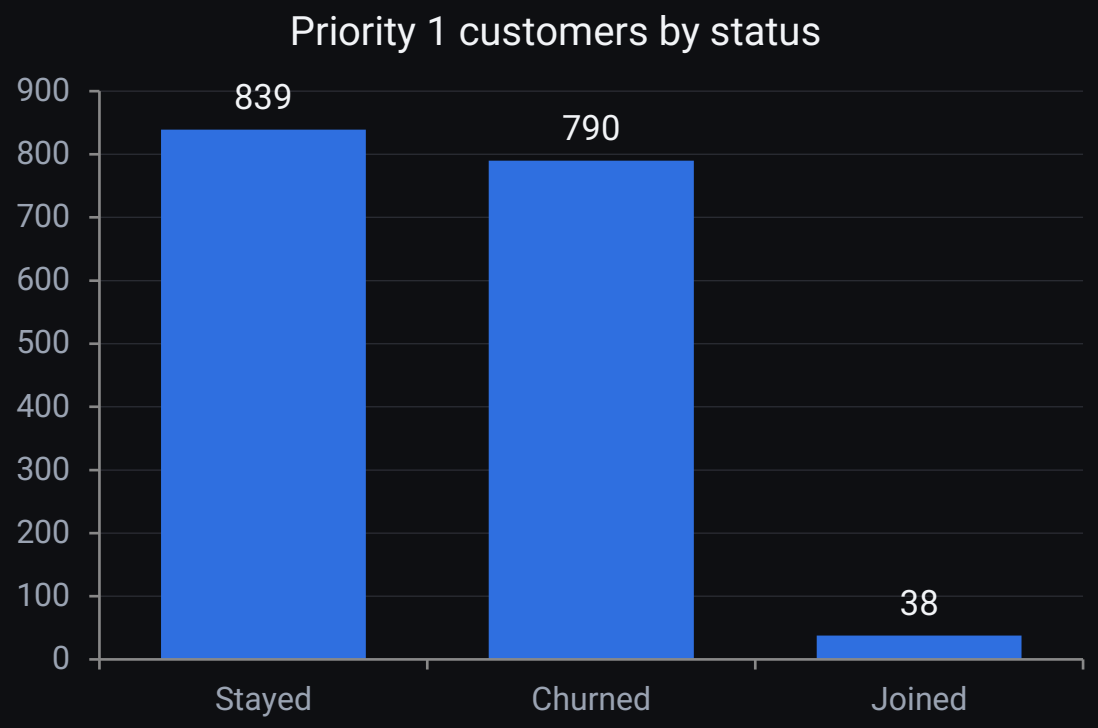
Q1: How many high-value customers are also at high risk, and what are their statuses?

Identifying & Segmenting Customers

1,667

Priority 1 customers (High Value – High Risk)

Status	Customers	Share
Stayed	**839**	**50.33%**
Churned	790	47.39%
Joined	38	2.28%



Key insight: the 839 active Priority 1 customers are the main retention target.

Q2: What customer value and monthly charges are exposed to churn risk?

Measuring Revenue at Risk

CLTV at risk
$4,567,572.00

MRR at risk
$55,498.65/mo

Avg. CLTV
$5,444.07

Avg. MRR
$66.15/mo

Key insight

Approximately $4.57 million in customer lifetime value and $55,500 in monthly charges are exposed to churn risk (839 active Priority 1 customers).

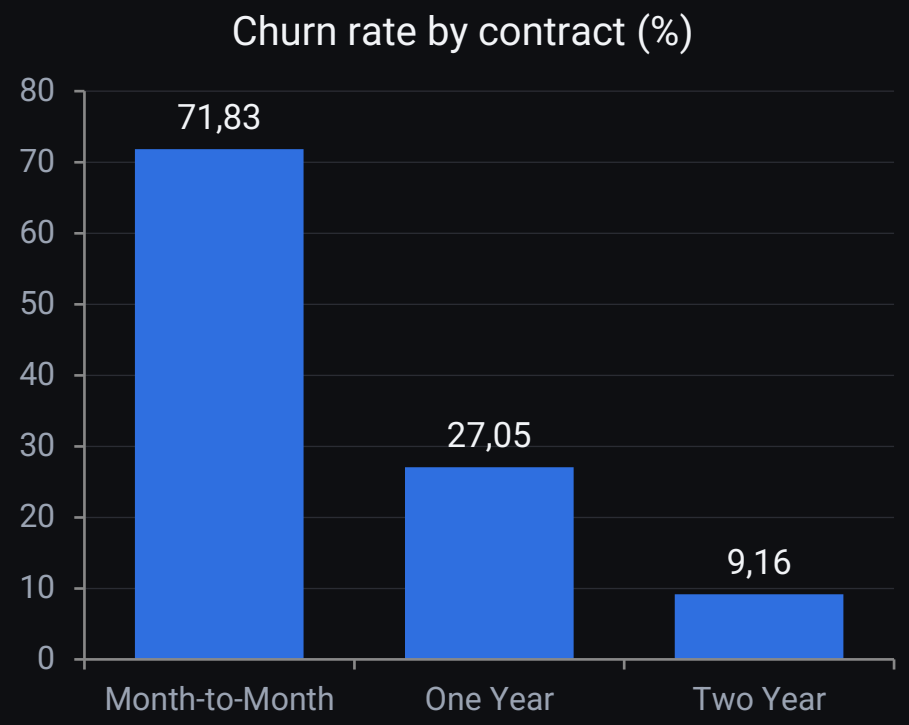
Potential exposure, not losses that have already occurred or are certain to happen.

Q3: Is contract type associated with churn among Priority 1 customers?

Churn Patterns by Contract Type

Contract	Customers	Churned	Churn rate
Month-to-Month	**923**	**663**	**71.83%**
One Year	329	89	27.05%
Two Year	415	38	9.16%

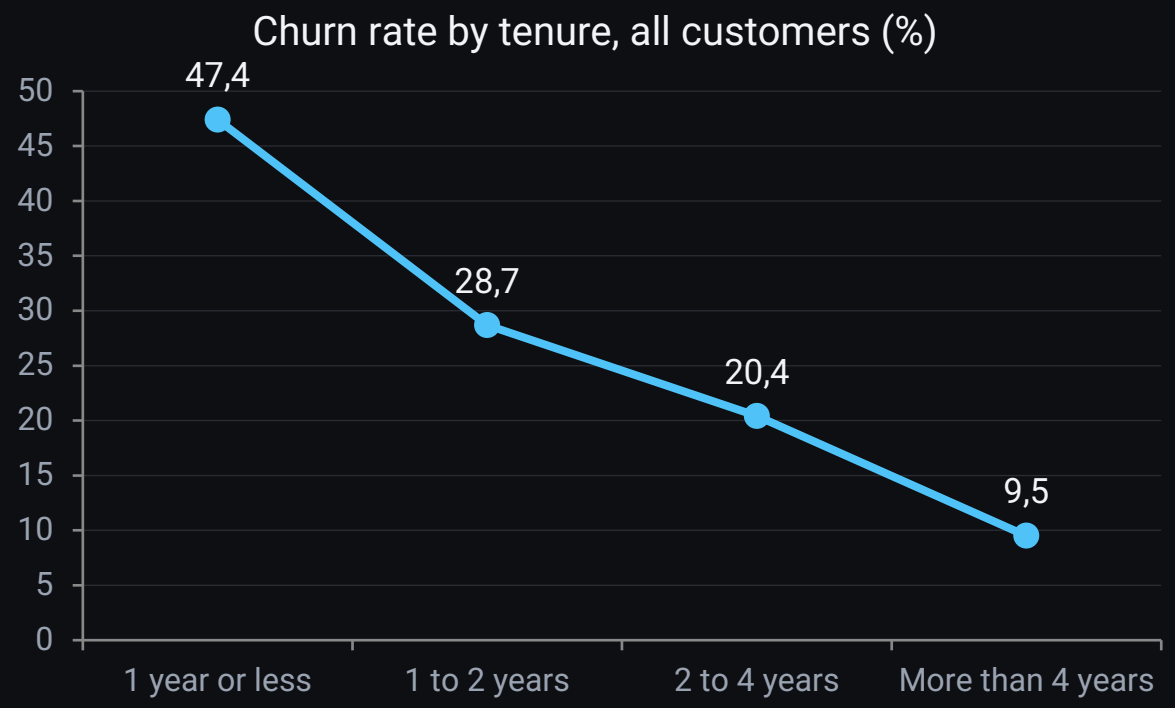
Offering longer-term contracts with carefully designed incentives is a potential retention strategy worth testing.



Month-to-month customers have the highest churn rate among Priority 1 customers.

Q3: How are subscription duration and customer satisfaction associated with churn?

Churn by Tenure & Satisfaction



Internet users, tenure ≤ 1 year

2.66 avg. satisfaction

No internet service, tenure ≤ 1 year

3.61 avg. satisfaction

The first year is critical: prioritize onboarding, early support and satisfaction monitoring.

Satisfaction figures are for customers with CLTV at or above the median. Patterns do not prove that tenure or internet service causes churn.

Q4: Which retention strategies should be used, and what is the simulated budget limit?

Retention Strategies & Budget

Retention target
839 customers

Total budget limit
$456,757.20

Avg. per customer
$544.41

Strategy (budget = 10% of CLTV)	Customers	Share
Loyalty Programs and Regular Monitoring	614	73.18%
Annual Contract Offers with Measured Incentives	225	26.82%

Offer longer-term contracts to month-to-month customers; give other eligible customers loyalty programs and monitoring. The budget is a CLTV-based simulation, not a proven optimum.

CONCLUSIONS

Conclusions & Business Recommendations

Key Findings

- Segmentation: focus on 839 active Priority 1 customers.
- Exposure: about $4.57 million CLTV and $55,500 monthly charges at risk.
- Patterns: month-to-month churn is 71.83%; first-year customers also churn most.
- Actions: test longer-term contracts, strengthen onboarding, follow up on satisfaction issues.

Final Recommendation

- Start with a pilot for high-value, high-risk customers.
- Retention rate: share of targeted customers who stay.
- Cost per retained customer.
- Protected margin: profit preserved by preventing churn.

Use pilot results to choose the strategies with the best business value, then adjust the program and budget.

THANKYOU

APPENDIX

APPENDIX

Feature Engineering & Customer Segmentation

Derived Features

- projected_annual_charges = monthly_charges × 12
- retention_budget_cap = cltv × 10%
- Source datasets: Status_Analysis.csv, Payment_Info.csv, Service_Options.csv, joined using customer_id

Tenure Group Classification

- New: ≤ 12 months
- Mid-term: > 12–24 months
- Established: > 24–48 months
- Loyal: > 48 months

Customer Tier Classification

Median thresholds: CLTV = $4,527 | Churn Score = 61

Customer Tier	Classification Rule
Priority 1	**High CLTV + High Churn Score**
Priority 2	High CLTV + Low Churn Score
Priority 3	Low CLTV + High Churn Score
Priority 4	Low CLTV + Low Churn Score

High: value ≥ median. Low: value < median.

APPENDIX

Metric Calculation & Financial Exposure

Churn Metrics

- Churn Rate = Churned Customers ÷ Total Customers × 100%
- Overall: 26.54%
- Month-to-Month: 45.84%
- One Year: 10.71%
- Two Year: 2.55%

Retention Budget Simulation

- Budget per customer = cltv × 10%
- Total Budget Limit = Sum of retention_budget_cap = $456,757.20
- Average per customer = Total ÷ 839 = $544.41

Financial Exposure

Filter: customer_tier = Priority 1 and customer_status = Stayed

Metric	Calculation	Result
Target Customers	Count of filtered customers	839
CLTV at Risk	Sum of cltv	$4,567,572
Average CLTV	CLTV at Risk ÷ 839	$5,444.07
MRR at Risk	Sum of monthly_charges	$55,498.65/month
Average MRR	MRR at Risk ÷ 839	$66.15/month

Note: exposure is the value associated with the selected group, not guaranteed losses.

APPENDIX

Satisfaction & Churn Analysis

Customer Satisfaction

Filter: cltv ≥ $4,527 and tenure ≤ 12 months. Average = Sum of satisfaction_score ÷ Number of Customers.

Internet Service	Average Satisfaction
Yes	2.66
No	3.61

Churn Category Distribution

Customers counted by churn_category among churned customers.

Churn Category	Customers
Competitor	**841**
Attitude	314
Dissatisfaction	303
Price	211
Other	200

Retention Strategy Rules

Condition	Recommended Action
Month-to-Month + Satisfaction ≤ 2	Follow up on complaints and consider annual-contract incentives
Month-to-Month only	Test incentives for longer-term contracts
Satisfaction ≤ 2 only	Investigate complaints and service quality
Other customers	Loyalty programs and periodic monitoring

Interpretation: satisfaction differences are descriptive, not causal. Retention actions are rule-based recommendations, not proven estimates of campaign effectiveness.

DASHBOARD

Power BI

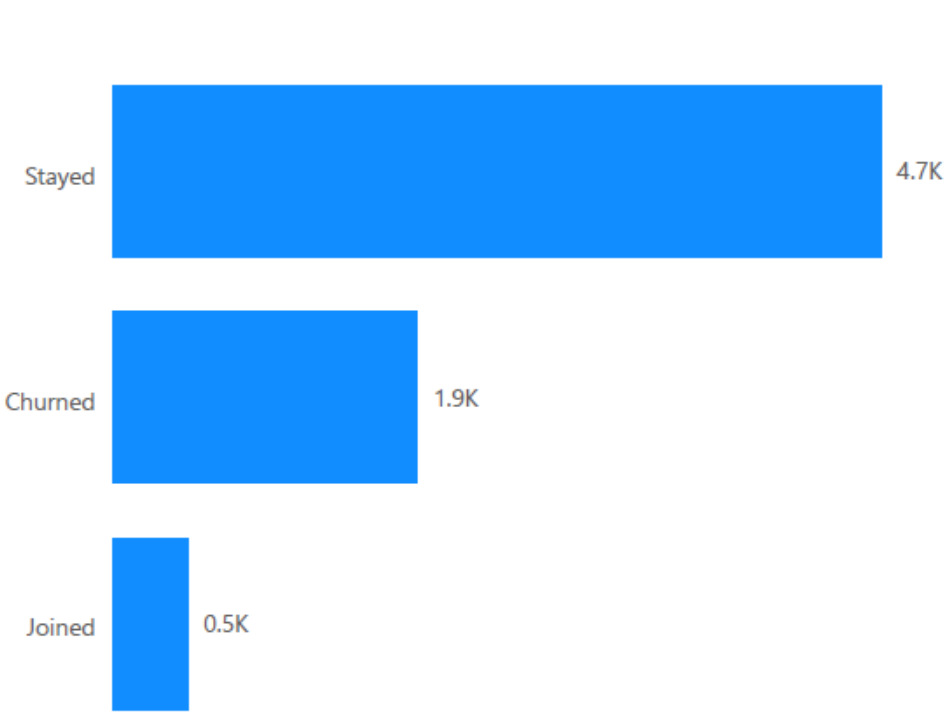
Customer Churn Overview

7.04K	26.54%	1.87K	4.72K	456.76K
Total Customers	Churn Rate	Churned Customers	Stayed Customers	Retention Budget Cap

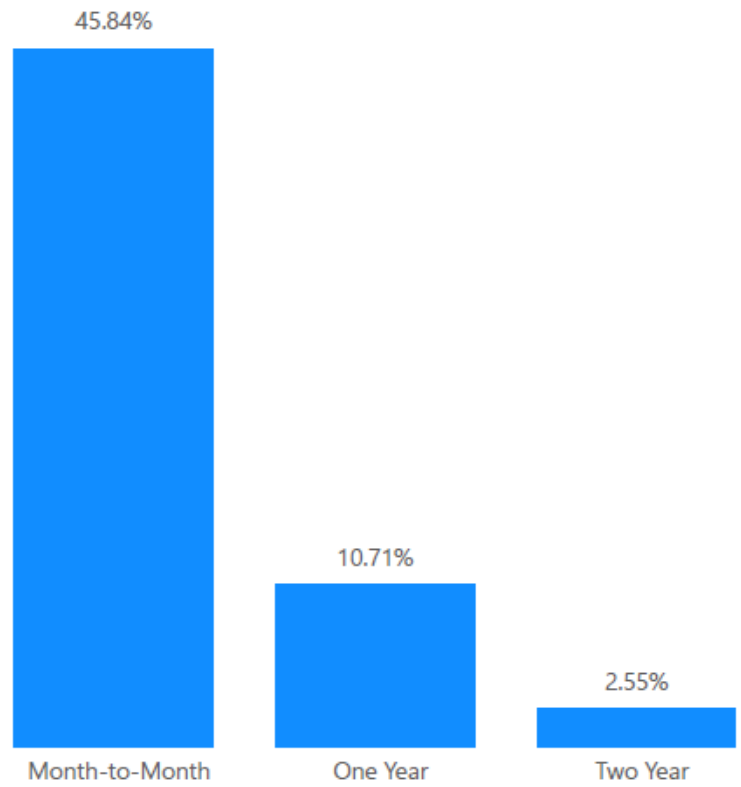
payment_method

All

Total Customers by customer_status



Churn Rate by contract



contract

- [ ] Month-to-Month
- [ ] One Year
- [ ] Two Year

internet_service

- [ ] No
- [ ] Yes

tenure_group

- [ ] 01. <= 1 Tahun (Baru)
- [ ] 02. 1 - 2 Tahun (Menengah)
- [ ] 03. 2 - 4 Tahun (Mapan)
- [ ] 04. > 4 Tahun (Loyal)

Customer Risk Segmentation

839
Priority 1 Active

5M
CLTV at Risk

55.50K
MRR at Risk



customer_tier
- [x] Priority 1 (High Value - High Risk)
- [ ] Priority 2 (High Value - Low Risk)
- [ ] Priority 3 (Low Value - High Risk)
- [ ] Priority 4 (Low Value - Low Risk)

customer_status
- [ ] Churned
- [ ] Joined
- [x] Stayed

customer_id	customer_status	cltv	churn_score	contract	tenure	satisfaction_score	internet_service	monthly_charges	retention_budget_cap
0003-MKNFE	Stayed	5414	66	Month-to-Month	9	5	Yes	59.90	541.40
0036-IHMOT	Stayed	5892	65	One Year	55	3	Yes	103.70	589.20
0052-YNYOT	Stayed	5462	72	One Year	67	5	No	20.55	546.20
0071-NDAFP	Stayed	5230	65	Two Year	25	3	No	25.50	523.00
0078-XZMHT	Stayed	6322	76	Two Year	72	3	Yes	85.15	632.20
0083-PIVIK	Stayed	4761	67	One Year	64	3	Yes	81.25	476.10
0103-CSITQ	Stayed	6026	65	One Year	57	5	Yes	109.40	602.60
0112-QAWRZ	Stayed	5560	66	Month-to-Month	16	3	Yes	90.80	556.00
0114-IGABW	Stayed	5224	76	Two Year	71	5	Yes	58.25	522.40
0141-YEAYS	Stayed	5990	77	Two Year	27	4	Yes	86.45	599.00
0142-GVYSN	Stayed	5712	68	Month-to-Month	26	5	No	20.30	571.20

contract
- [ ] Month-to-Month
- [ ] One Year
- [ ] Two Year

Clear all slicers

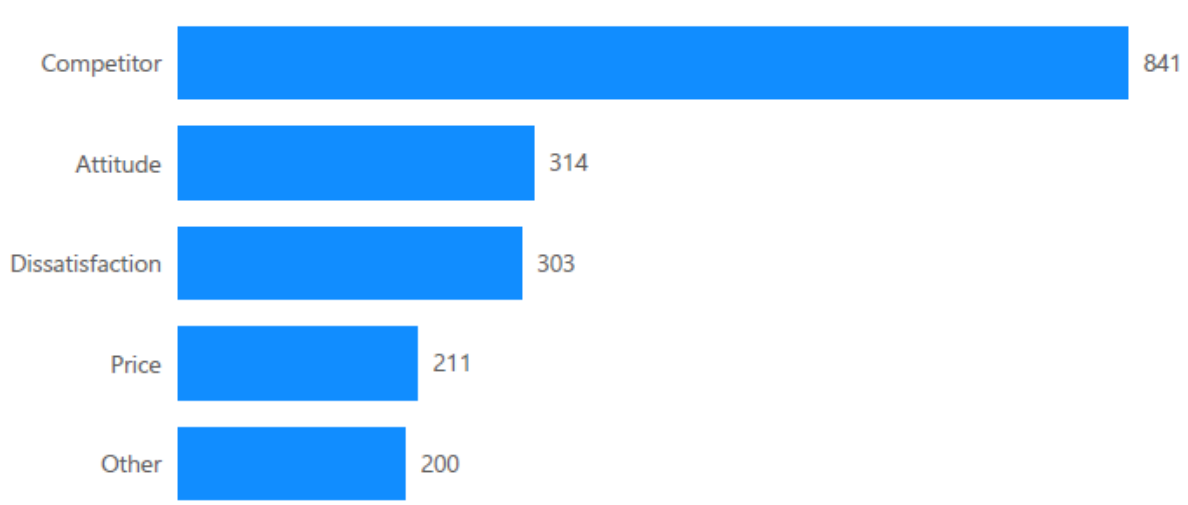
Retention Strategy & Financial Impact

customer_tier: All

contract: All

churn_category: All

Churned Customers by churn_category



Findings or Conditions	Recommendations
Churn reason related to competitors	Review pricing, service packages, and value propositions
High value, high risk, and still active	Prioritize personalized retention programs
Low customer satisfaction score	Prioritize investigating and resolving customer complaints
Lower value, high risk	Test low-cost retention interventions
Month-to-month contract	Test incentives to encourage longer-term contracts

Churned Customers by churn_reason

